



Nontarget pesticide impacts on pest natural enemies: progress and gaps in current knowledge

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Avoiding pesticide nontarget effects on natural enemies is a cornerstone of conservation biological control. Recent advances in this field have included increased examination of nuanced sublethal effects, including microbiome changes. There is an interest in lifetable-based approaches, while also simplifying results to reduce the amount of information a grower needs to interpret to make a judicious application decision. Newer pesticides are showing promise for selectivity to both natural enemies and humans. Major research gaps still remain, with few published studies on ground-dwelling natural enemies, herbicides, adjuvants, or pesticide mixes. Translating the results of laboratory assays to field-level effects remains a major challenge. Field studies examining entire management programs and meta-analyses of laboratory studies may begin to address this issue.

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Introduction

Conservation biological control is an important component of integrated pest management in agriculture. Within this tactic, avoiding pesticide nontarget effects through selective pesticide applications is the critical first step [1]. Without first avoiding pesticide applications that are harmful to key natural enemies, other conservation strategies, such as habitat provisioning, are often doomed to fail [2,3]. In order to provide growers with pesticide-selectivity recommendations, researchers conduct trials in the lab and field to determine which pesticides are the least disruptive to biological control.

In this paper, I overview recent developments in the field of pesticide nontarget effect research as it pertains to natural enemies of agricultural pests. This review discusses the increased focus on sublethal effects (Figure 1) and selectivity of newer pesticides. I indicate critical gaps in the literature, emphasizing natural enemy groups and pesticides that are understudied. Ongoing difficulties translating lab results to field-level effects are highlighted, using research conducted in northwestern U.S. orchards as an example. Finally, I suggest meta-analysis as a tool for generalizing across diverse nontarget effect studies.

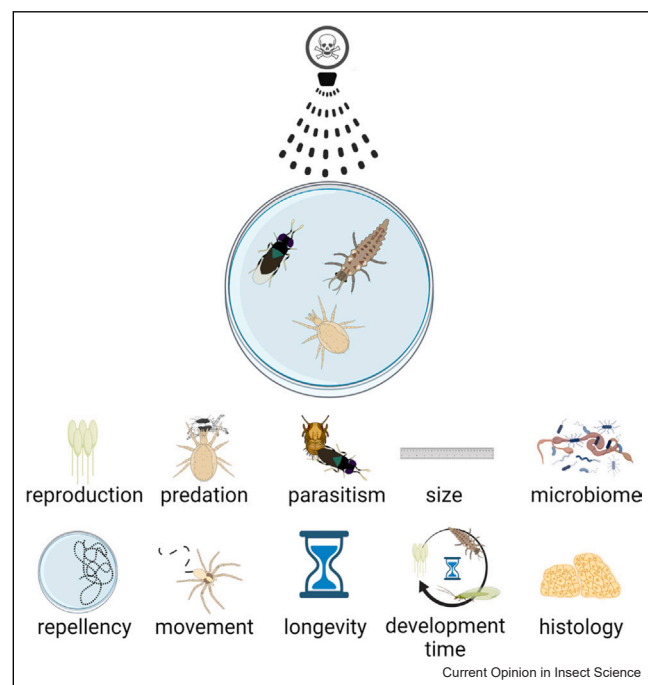
Sublethal effects

Sublethal effect testing is becoming increasingly common [4]. There are several recent studies on transgenerational effects of pesticide exposure, with researchers demonstrating that unexposed offspring of exposed females have reduced reproductive capability and survival, and increased development time; the natural enemies were exposed to dry residues and included a laelapid (various insecticides) [5], ladybeetles (imidacloprid, sulfoxaflor, and tolfenpyrad) [6,7], and an anthocorid (acetamiprid and imidacloprid) [8]. One approach, the lifetable selectivity index, uses percent reduction in live offspring as a cumulative measure of adult mortality, fecundity changes, egg hatch, and juvenile mortality after hatch and compares this percent reduction in the pest versus the predator [9,10]. These methods improve upon previous studies in not only in just gathering additional information, but also in that they consider how exposure of an individual might then affect population growth.

There is increased interest in how pesticides affect natural enemy movement, which is aided by improved motion-tracking cameras and software [11,12]. These studies can also be used to examine repellency, which may be important in understanding the differences in outcomes between laboratory and field studies; a repellent material in the field may cause a natural enemy to seek untreated refugia, minimizing contact with spray residues [13]. Studies of natural enemy movement can also identify how sublethal effects alter predator responses to volatiles of pests [14] and host plants [15]. Other studies are finding that sublethal concentrations of pesticides can cause histological changes in the digestive tract [16] and alterations to the microbiome [17,18].

Pesticide sublethal doses can sometimes increase reproduction. Hormesis is infrequently reported in natural

Figure 1



Laboratory studies on sublethal effects examine how pesticides can alter various parameters of natural enemy life history and behavior.

enemies [19], potentially because nontarget effect screening focuses on (often-lethal) field doses [20]. Pyrethroids can induce hormesis in predatory stink bugs [21,22]. Hormesis was recently observed in *Podisus maculiventris* exposed to imidacloprid [23,24] and in a predatory mite exposed to hexythiazox [9]. Pyriproxyfen increased egg production and female–female aggressiveness in a parasitoid without affecting survival or host-seeking [25]. Spirotetramat exposure decreased host location and handling times in *Encarsia formosa* [26]. Hormesis may even be exploited for increasing reproduction of mass-reared natural enemies [27], but this has not yet been explored.

Selectivity of newer pesticides

Selective conventional insecticides include the diamides and selective miticides that are marketed as harmless to natural enemies. While these products are much less harmful to natural enemies than broad-spectrum insecticides, no selective pesticide is ‘safe’ for all natural enemies [9,20]. For instance, the diamide chlorantraniloprole caused high mortality in *Cycloneda sanguinea* [28]. Miticides marketed as selective to phytoseiids have been found to be harmful to some species of phytoseiids in laboratory tests and the field [9,29,30].

Because the mode of action of RNA interference (RNAi)-based insecticides is sequence-guided, these

newer insecticides show great promise for selectivity. In a thorough study of potential nontarget effects, there was no alteration in gene expression, enzyme levels, survival, or parasitism rates in *Telenomus podisi* after exposure to dsRNA that is lethal to its host, *Euschistus heros* [31]. The authors suggest using their study as a map for examining potential hazards of RNAi-based insecticides through both traditional assays and bioinformatics approaches.

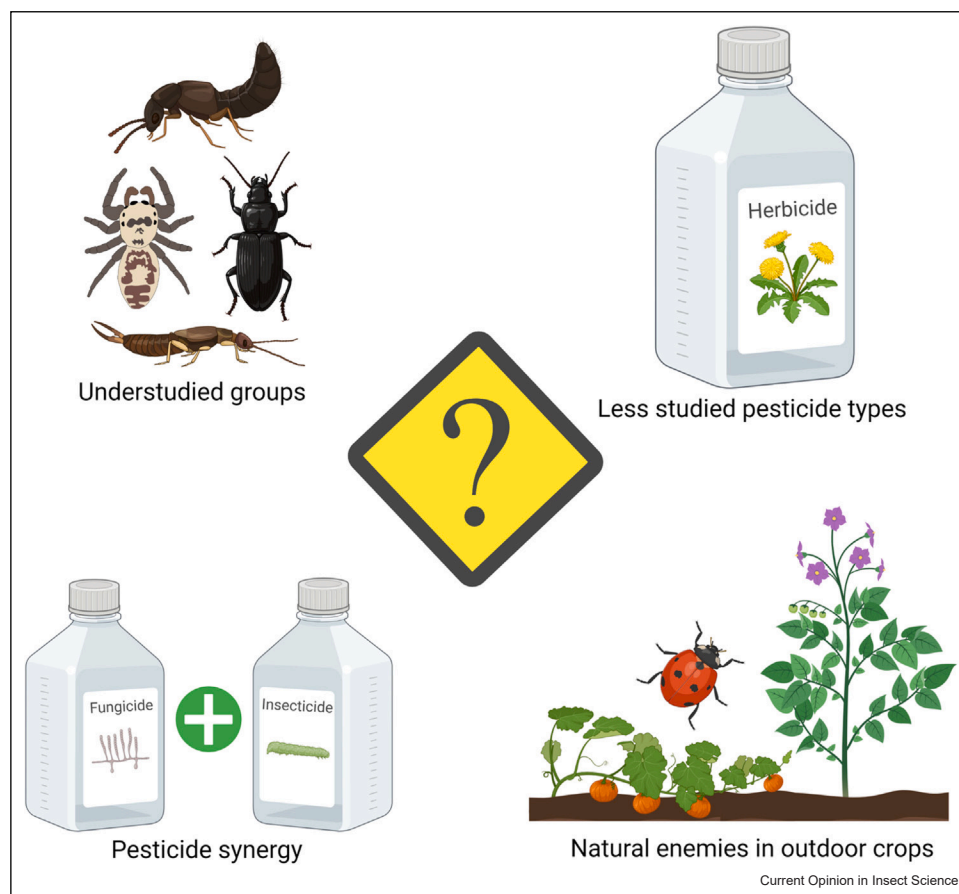
Some food additive products have insecticidal properties and have been examined for their nontarget effects. Erythritol is a non-nutritive sweetener that has potential for managing a variety of pests and has been found to be minimally harmful to a parasitoid wasp and a predatory mite [32,33]. Methyl benzoate, a food-flavoring agent derived from apple juice, has insecticidal activity against several pest species and few nontarget effects on predatory hemipterans [34,35].

Research gaps

There are both pesticide types and natural enemy groups that are understudied (Figure 2). There are particularly few studies on herbicide effects [11,20,36,37]. The physiological mechanisms of herbicides on arthropod natural enemies are virtually undescribed [11,18,36,38]. Pendimethalin exposure altered gut microbiota in a carabid, especially microbiota related to carbohydrate, amino acid, and lipid metabolism [17]. Glyphosate application changed the endosymbiont community composition in *Harmonia axyridis* and caused reduced body weight [18]. Herbicides can alter movement and predation rates of spiders [11,39]. Field studies on herbicide effects on natural enemies are exceedingly rare. Paraquat applications have been associated with pest mite outbreaks due to nontarget effects on predatory mites [40,41].

Adjuvants can increase pesticide toxicity to natural enemies. Toxicity of imidacloprid to the lacewing *Chrysoperla nipponensis* was increased when imidacloprid was mixed with an organosilicone surfactant [42]. The mixture resulted in a lower LC₅₀, slowed development, and decreased longevity and reproduction compared with imidacloprid alone [42]. Various surfactants were shown to decrease predatory activity of *Pardosa* spiders, either alone or in combination with a formulated glyphosate product [43,44]. The glyphosate product alone did not alter spider behavior [43]. However, these studies are among the few that test nontarget effects of adjuvants alone or in combination with pesticides on natural enemies. Synergistic effects of pesticide mixes, such as fungicide and insecticide combinations, are becoming more commonly tested in bees but remain mostly untested in natural enemies. More studies are also needed on natural enemy pesticide exposure via seed treatments [45].

Figure 2



Several key areas are understudied in pesticide nontarget effect research.

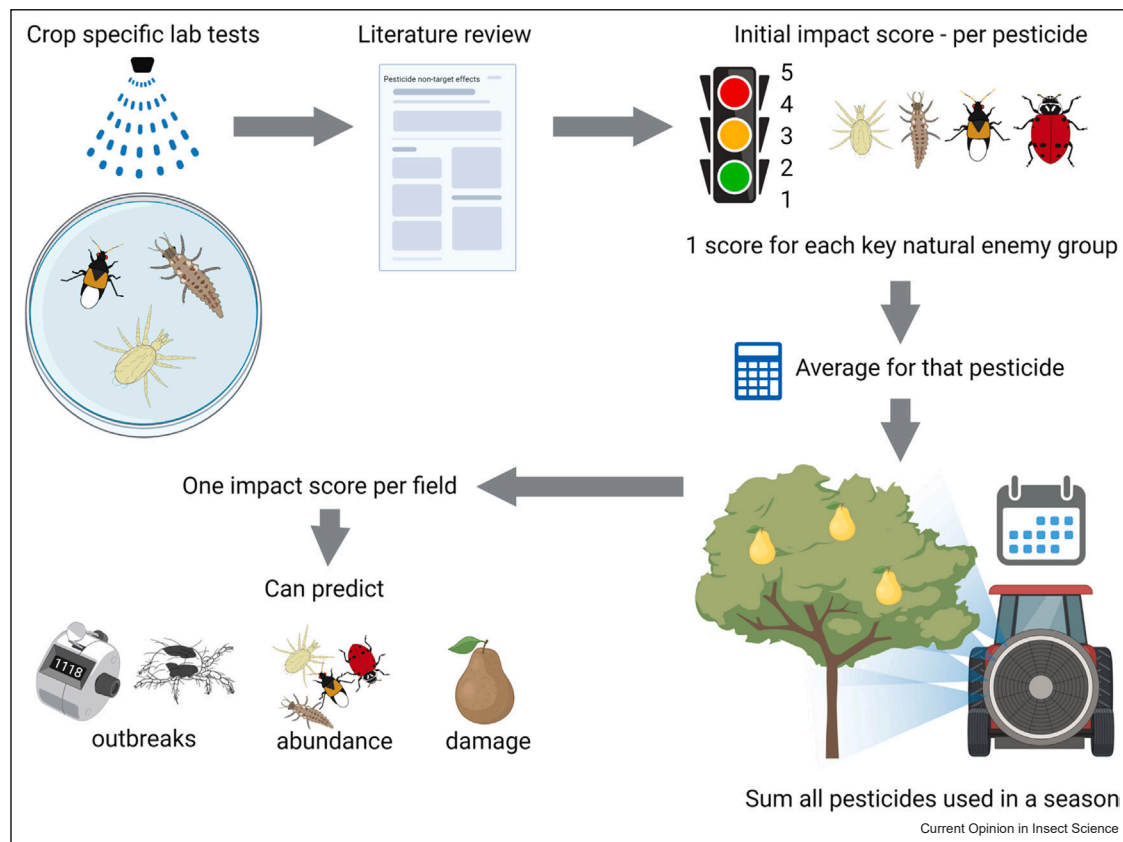
Pesticide nontarget research on ground-dwelling natural enemies is scarce, with ants, carabids, staphylinids, nonphytoseiid mites, and spiders under-represented in the literature [11,45,46]. Syrphids are also infrequently studied [45]. In general, nontarget effects on natural enemies outside of Europe and Canada/the United States are rarely researched [45]. Finally, due to the easy accessibility of commercially reared natural enemies for greenhouses, the majority of nontarget effect testing has been conducted on this group [20,45]. This leaves pesticide effects on natural enemies from unprotected cropping systems less understood.

Translating results from lab to field

The goal of laboratory testing for nontarget effects is to inform growers on how to minimize disruption to biological control in their fields. However, translating laboratory results to field settings remains an ongoing struggle. LC_{50} -based approaches are limited because they do not test field rates, sublethal effects, or multiple life stages [10]. The International Organization for Biological Control — West Palearctic Regional Section

Working Group (IOBC) has established protocols for testing pesticide effects on a list of model natural enemies, and has guidelines for rating pesticide toxicity (1–4 scale), which are documented in its database [47]. This system allows for clear comparisons between studies that follow these protocols and lab-based IOBC ratings have been correlated to field effects [48]. However, many researchers do not use these guidelines, potentially because they are not always relevant to the biology of the natural enemy tested or may be impractical in specific cases [45]. The IOBC-representative organisms are limited in taxonomic diversity and may disadvantage geographic regions that have important natural enemies that are not represented in the database [45]. Furthermore, the database does not always report explicit data (e.g. percent mortality), with only the rating provided in some instances, making it difficult for other researchers to interpret the results in depth [20,45]. Other approaches, such as measuring changes in demographic parameters (e.g. intrinsic rate of increase, net reproductive rate, and generation time) are useful [49], but are difficult to translate directly to growers. Approaches

Figure 3



The process of determining natural enemy impact scores for a field.

that compare toxicity of a pesticide to the pest and its natural enemy may be useful in predicting outbreaks and also incorporate the potential of pest resistance when determining selectivity [9,10,36,50], however, these are less commonly used.

Field studies alone can also be difficult to draw conclusions from. Reductions in natural enemy populations after a pesticide application may be due to reduction in prey availability (indicating an effective pesticide) or changes in resources impacted by pest density (honeydew, host plant quality), rather than direct harm to the natural enemies [49,51]. Researchers can counteract this by making additional maintenance sprays to control the target pest, but these are not always effective, due to the many biotic and abiotic factors that can affect insecticide efficacy [51]. Field studies should use predator:prey ratios (as opposed to just changes in predator abundance) to also account for this issue and to determine if a pesticide favors biological control [51,52].

Recent work in northwestern U.S. orchards has examined differences in outcomes between lab and field

nontarget studies. This project found that while assays could sometimes predict secondary pest outbreaks (more harmful pesticides to natural enemies, more likely to see an outbreak), this did not usually correspond with changes in natural enemy populations in the field; plots treated with the more harmful insecticides (according to the lab assays) did not usually have reduced populations of natural enemies [53,54]. Surprisingly, the exception to this was earwigs, where the lab assays accurately predicted which pesticides would decrease earwig field abundance and use of these pesticides correlated with secondary pest outbreaks [53,54]. There were no 'false negatives', pesticides deemed 'harmless' in laboratory trials did not reduce natural enemy abundance or induce pests [53,54]. The researchers also noted that difficulty with finding differences between treatments could be due to using a selective insecticide (chlorantraniliprole) as the 'control' because most grower participants were unwilling to leave untreated control plots [53]. This will continue to be a challenge in nontarget research because large-plot farm-scale studies are needed to see realistic field effects, especially for highly mobile natural enemies that easily move between plots.

A newer approach may address this issue (Figure 3). The annual Washington State University Crop Production Guide [55] gives pesticides a natural enemy impact score [56,57]. These scores are based on a literature review of various lab studies [56,57]. A single score is assigned to an orchard for a year and accounts for what pesticides are applied and number of applications [56,57] (Figure 3). This may be the most realistic approach for pesticide nontarget effect measurement in the field, as it assesses an entire management program. Unlike the prior method (that examined individual pesticides) [53,54], these scores predicted natural enemy abundance and higher populations of natural enemies in early summer and at harvest were associated with improved fruit quality [57]. Impact scores are useful to growers because they reduce the amount of information they need to assess: one score per pesticide instead of many nontarget effects for multiple natural enemies. The natural enemy impact score has been used in other studies where it was found that the score was a better predictor of natural enemy abundance than landscape effects [58]. Higher impact scores were also associated with pest mite outbreaks, indicating that the scores can predict biological control disruption [58], unlike field studies focusing on individual pesticides [53,54]. Because the scores rely on laboratory studies, ongoing nontarget assays will still be important to keep the database used to calculate the scores updated. Other studies have also found utility in focusing on whole-program spray regimes to assess nontarget effects on natural enemies [59]. This may be the next common approach to nontarget effects: first examine a series of pesticides in the lab for a variety of natural enemies, then assess the impact of spray programs in the field using the lab data to make predictions based on a simplified metric.

Identification of measures that streamline the lab testing process is critical, because there is a virtually infinite number of combinations of species, life stages, formulations, exposure methods, and residue ages to test. Systematic reviews of nontarget effects help practitioners make evidence-based generalizations. The most complete systematic review of nontarget effects was conducted in the 1980s [50]. Systematic reviews since then have been more narrowly focused, covering a single-arthropod species, group, or pesticide type [36,46], or a cropping system [37,45]. While systematic reviews are useful, they are still difficult to generalize from and are essentially a form of ‘vote-counting’, drawing a conclusion based on the number of studies showing an effect [60,61]. To date, only two meta-analyses on pesticide nontarget effects on natural enemies have been published, one on *Trichogramma pretiosum* [62] and one on phytoseiids [20]. These analyses were useful in drawing broader conclusions about pesticide impacts on these natural enemy groups. Additionally, the data extracted for meta-analyses can then be made publicly

available, essentially creating open-source databases of nontarget research. Going forward, meta-analyses will be useful to researchers wanting to compare different pesticides, formulations, natural enemy species, methods of exposure, and a variety of other questions that can be addressed using data standardized across many studies.

Conclusion

Significant headway has been made in the field of pesticide nontarget effects on natural enemies. There has been an increased emphasis on sublethal effects, including intergenerational impacts, and the ability to measure various physiological changes has improved. Newer modes of action show promise for increasing pesticide selectivity. However, substantial gaps in knowledge remain, with more research needed on the impacts of herbicides, adjuvants, and pesticide mixtures. It will also be a continuing challenge to correlate laboratory results to field and economic effects, such as yield loss. The impact score method shows promise and may translate well to cropping systems outside of orchards.

Terms definitions box

Useful terms

Adjuvant	A product or ingredient that is designed to enhance the effect of pesticide. This category includes surfactants, which aid in breaking surface tension and help the pesticide stick to the plant.
Broad-spectrum insecticide	Insecticides that have toxic activity against a variety of taxonomically diverse pests. They are usually harmful to many natural enemy groups.
Conservation biological control	Practices used to maintain or attract resident natural enemies into a crop. Includes practices that enhance the landscape (e.g. floral resource plantings), but also practices that minimize ecological disturbances (e.g. no-till, selective pesticide use).
Conventional pesticide	Pesticides that are not certified organic. This category excludes microbial pesticides, plant oil extracts, and soaps, among others.
Hormesis	A response to a moderate stressor that is typically adaptive to the exposed organism. In the context of pesticides, this typically refers to a lower (nonlethal) dose of pesticide resulting in what appears to be a beneficial response, often increased reproduction.
LC ₅₀	Concentration of a substance that kills 50% of the tested population.
Natural enemy	In agriculture, this typically refers to predators, parasitoids, and parasites of crop pests.
Pesticide selectivity	The degree to which a pesticide is only harmful to the target pest. In the context of biological control, this refers to comparisons of pesticide toxicity to a pest and its main natural enemy or enemies.
Residue	In the context of nontarget effect research, residues refer to either fresh (wet) or dry pesticide in the environment and are a type of

(continued)

Useful terms

Sublethal effect	exposure. Other exposure types include direct spray/direct contact and feeding of contaminated prey or other food items. Side effects of a pesticide on individuals that survive exposure. Examples are provided in Figure 1.
Transgenerational effect	Effects of pesticides that are observed in the descendants of exposed individuals.

Data Availability

No data were used for the research described in the article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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